

WASP-80b

R-2108

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-80_190716 · created 2026-08-11T00:12:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458680.9424 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.068
Transit depth (Rp/Rs)^2	2.48 +/- 2.13 [%]
Orbital Inclination (inc)	84.92 +/- 1.75
Airmass coefficient 1 (a1)	1.1118375 +/- 4e-06
Airmass coefficient 2 (a2)	-0.055 +/- 0.061
Scatter in the residuals of the lightcurve fit is	2.5e-05 %
Best Comparison Star	#3 - [391, 262]
Optimal Aperture	104.02
Optimal Annulus	189.52
Transit Duration (day)	0.00013 +/- 0.00049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.068 vs 0.1714 (deviation 0.21 σ)
Duration fit vs published	0.00013 d vs 0.0892 d (deviation 181.71 σ)
Mid-time O-C	4.5 min
Depth SNR	2.3
Residual scatter	2.5 %

Light curve

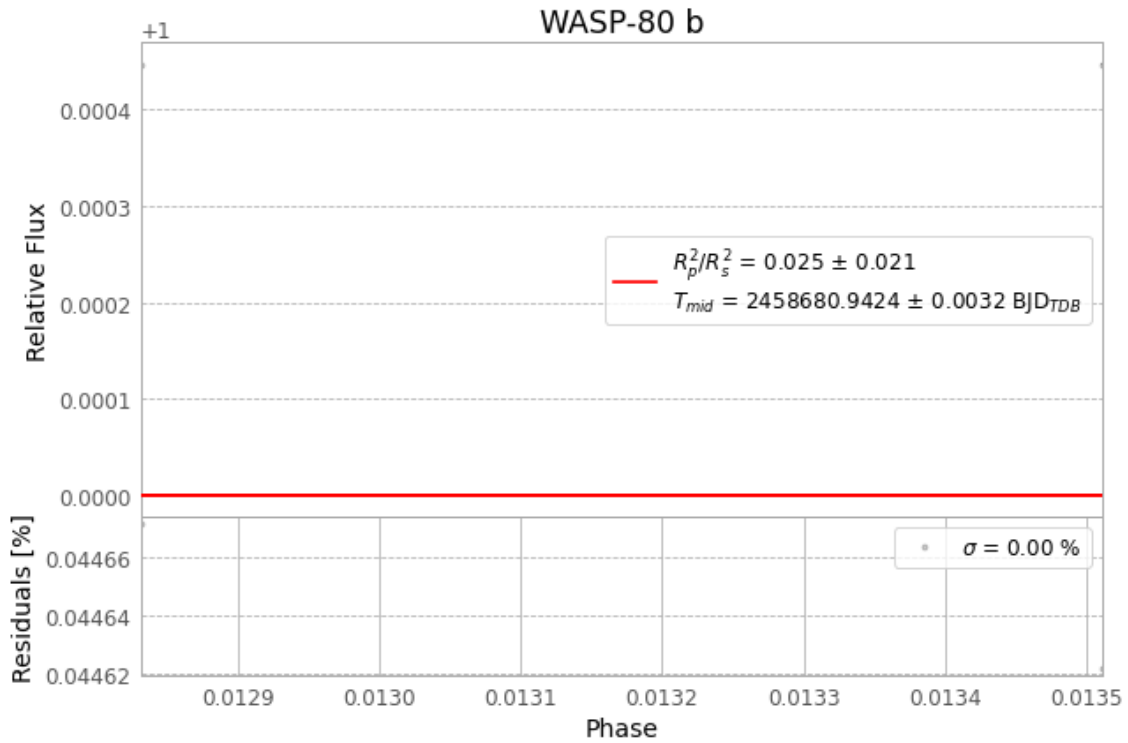


Figure 1 — FinalLightCurve_WASP-80 b_2019-07-16.png (SHA-256 07a6bd4251000d7e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [599, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b17feae5dda08fbd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

45 FITS frames (30 MB), WASP-80190716091512.FITS → WASP-80190716112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-190716.log (6a4543eb2c65...), FinalLightCurve_WASP-80 b_2019-07-16.png (07a6bd425100...), FinalLightCurve_WASP-80 b_2019-07-16.csv (43295b00d8e4...)

Compute resources

Wall time 952.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:12Z.